CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (CBIT)

Department Of Computer Science And Engineering

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CASE STUDY REPORT

Title: Wine Quality Assessment Using Physicochemical Properties

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ABSTRACT
This case study investigates the relationship between physicochemical properties and wine quality using statistical analysis, exploratory data analysis and multiple linear regression.
The main objectives are:
- To analyse the statistical properties of alcohol, volatile acidity and residual sugar.
- To examine relationships between physicochemical measurements and wine quality.
- To visualise quality-score distributions and attribute correlations.
- To build a regression model that predicts wine quality.
- To evaluate the model using MAE, MSE, RMSE and R-squared.
- To recommend laboratory measurements that can support quality screening.

Main findings:
- Residual sugar has the greatest relative variability among the three selected attributes.
- Alcohol has the strongest positive Pearson correlation with quality.
- Volatile acidity has the strongest negative Pearson correlation with quality.
- The regression model achieves an R-squared value of approximately 0.403 on the test set.
- Laboratory-based predictions can support quality assessment but should not completely replace sensory evaluation.

1. INTRODUCTION
Wine quality assessment is important for maintaining product consistency and customer satisfaction.
Traditional wine assessment often relies on sensory evaluation, which examines characteristics such as taste, aroma and overall quality.
Physicochemical measurements provide numerical information that can help explain differences in wine quality.
This case study uses data analysis to investigate whether laboratory measurements can support wine quality assessment.

2. PROBLEM STATEMENT
The company currently relies on sensory quality assessments. Management wants to investigate whether physicochemical measurements can help explain differences in wine quality and support a data-driven assessment process.
The main problem is to:
- Identify the physicochemical attributes most strongly associated with wine quality.
- Develop a regression model using the available laboratory measurements.
- Evaluate the accuracy of the model on unseen test data.

- Determine whether the model can be used for laboratory-based quality screening.

3. OBJECTIVES

The objectives of the case study are:

1. Calculate mean, median, mode, standard deviation and skewness for alcohol, volatile acidity and residual sugar.
2. Identify the attribute with the greatest relative variability using the coefficient of variation.
3. Calculate and interpret the Pearson correlation between alcohol and quality.
4. Generate a histogram, box plot and correlation heatmap.
5. Identify the two attributes most strongly associated with quality.
6. Develop a multiple linear regression model using 11 physicochemical attributes.
7. Split the data into training and testing sets using an 80:20 ratio.
8. Evaluate the model using MAE, MSE, RMSE and R-squared.
9. Identify influential predictors using standardized regression coefficients.
10. Examine multicollinearity using Variance Inflation Factor (VIF).
11. Recommend a suitable approach for laboratory-based quality screening.

4. DATASET DESCRIPTION

4.1 Dataset Overview

The case study ZIP file contains separate red wine and white wine datasets. The red wine dataset was selected for the main analysis.

Item	Description
Dataset used	Red wine quality dataset
Number of observations	1,599
Number of physicochemical attributes	11
Target variable	Quality
Data format	CSV
Analysis environment	Google Colab
Programming language	Python

4.2 Input Attributes

The 11 physicochemical attributes used as predictors are:

1. Fixed acidity
2. Volatile acidity
3. Citric acid
4. Residual sugar
5. Chlorides
6. Free sulfur dioxide

7. Total sulfur dioxide
8. Density
9. pH
10. Sulphates
11. Alcohol

4.3 Target Variable

- **Target variable:** Quality
- It represents the assigned quality score for each wine sample.
- The quality score is treated as a numerical target in the regression model.
- **4.4 Data Preparation**

The following steps were performed:

- Uploaded the ZIP file into Google Colab.
- Extracted the dataset files.
- Loaded the red wine CSV file using pandas.
- Used a semicolon delimiter to read the CSV correctly.
- Inspected the dataset dimensions and column names.
- Separated the 11 predictor attributes from the quality target.
- Divided the dataset into training and testing sets for modelling.

Data quality observation: The red wine dataset contains 240 exact duplicate rows. This needs to be considered when interpreting the model's evaluation results.

5. TOOLS AND TECHNOLOGIES USED

Tool / Library	Purpose
Google Colab	Running Python code online
Python	Data analysis and modelling
Pandas	Loading and analysing data
NumPy	Numerical calculations
Matplotlib	Creating graphs
Seaborn	Creating the correlation heatmap
Scikit-learn	Regression, data splitting and evaluation
Statsmodels	Multicollinearity analysis using VIF

6. METHODOLOGY

The analysis was completed in four stages.

Stage 1: Statistical Analysis

- Calculated descriptive statistics for the three selected attributes.
- Compared relative variability using the coefficient of variation.
- Calculated Pearson correlation between alcohol and quality.

Stage 2: Exploratory Data Analysis

- Generated a histogram of quality scores.
- Created a box plot of alcohol grouped by quality score.
- Generated a correlation heatmap.

- Identified the two strongest correlations with quality.

Stage 3: Predictive Modelling

- Selected all 11 physicochemical attributes as independent variables.
- Selected quality as the dependent variable.
- Split the dataset into 80% training data and 20% test data.
- Used random_state=42 for reproducibility.
- Trained a multiple linear regression model.
- Calculated MAE, MSE, RMSE and R-squared.
- Examined standardized coefficients and VIF values.

Stage 4: Business Decision-Making

- Recommended two measurements for quality screening.
- Evaluated whether the model was reliable enough to replace sensory evaluation.
- Identified limitations of the dataset and modelling approach.

7. TASK 1 — STATISTICAL ANALYSIS

7.1 Task 1(a): Descriptive Statistics

The following statistics were calculated for alcohol, volatile acidity and residual sugar:

- **Mean:** Average value of an attribute.
- **Median:** Middle value when observations are arranged in order.
- **Mode:** Most frequently occurring value.
- **Standard deviation:** Measures the spread of observations around the mean.
- **Skewness:** Measures the asymmetry of a distribution.

Table 1: Descriptive Statistics

Attribute	Mean	Median	Mode	Standard Deviation	Skewness
Alcohol	10.4230	10.2000	9.5000	1.0657	0.8608
Volatile Acidity	0.5278	0.5200	0.6000	0.1791	0.6716
Residual Sugar	2.5388	2.2000	2.0000	1.4099	4.5407

Interpretation

Alcohol

- Mean: approximately 10.423.
- Median: 10.2.
- Skewness is positive, indicating a distribution with a right-hand tail.

Volatile Acidity

- Mean: approximately 0.528.
- Median: 0.52.
- Positive skewness indicates some observations extend towards higher values.

Residual Sugar

- Mean: approximately 2.539.
- Median: 2.2.
- High positive skewness indicates a strongly right-skewed distribution.
- A relatively small number of observations have high residual sugar values.

TASK 1(a): DESCRIPTIVE STATISTICS

	Attribute	Mean	Median	Mode	Standard Deviation	Skewness	Coefficient of Variation (%)
0	alcohol	10.4230	10.20	9.5	1.0657	0.8608	10.2242
1	volatile acidity	0.5278	0.52	0.6	0.1791	0.6716	33.9244
2	residual sugar	2.5388	2.20	2.0	1.4099	4.5407	55.5351

7.2 Task 1(b): Greatest Relative Variability

The coefficient of variation (CV) was used to compare the relative variability of the three attributes.

Formula:

CV (%) = (Standard Deviation / Mean) × 100

Table 2: Coefficient of Variation

Attribute	Coefficient of Variation
Alcohol	10.22%
Volatile Acidity	33.92%
Residual Sugar	55.54%

Interpretation

- Alcohol has a coefficient of variation of approximately 10.22%.
- Volatile acidity has a coefficient of variation of approximately 33.92%.
- Residual sugar has the highest coefficient of variation at approximately 55.54%.

Conclusion: Residual sugar has the greatest relative variability among the three attributes.

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TASK 1(b): RELATIVE VARIABILITY

alcohol: CV = 10.22%
volatile acidity: CV = 33.92%
residual sugar: CV = 55.54%

Attribute with the greatest relative variability:
residual sugar
Coefficient of Variation = 55.54%
```

7.3 Task 1(c): Pearson Correlation Between Alcohol and Quality

Pearson's correlation coefficient measures the direction and strength of a linear relationship between two numerical variables.

Calculated result:

- Pearson correlation coefficient: **r = +0.4762**
- Direction: Positive
- Strength: Moderate

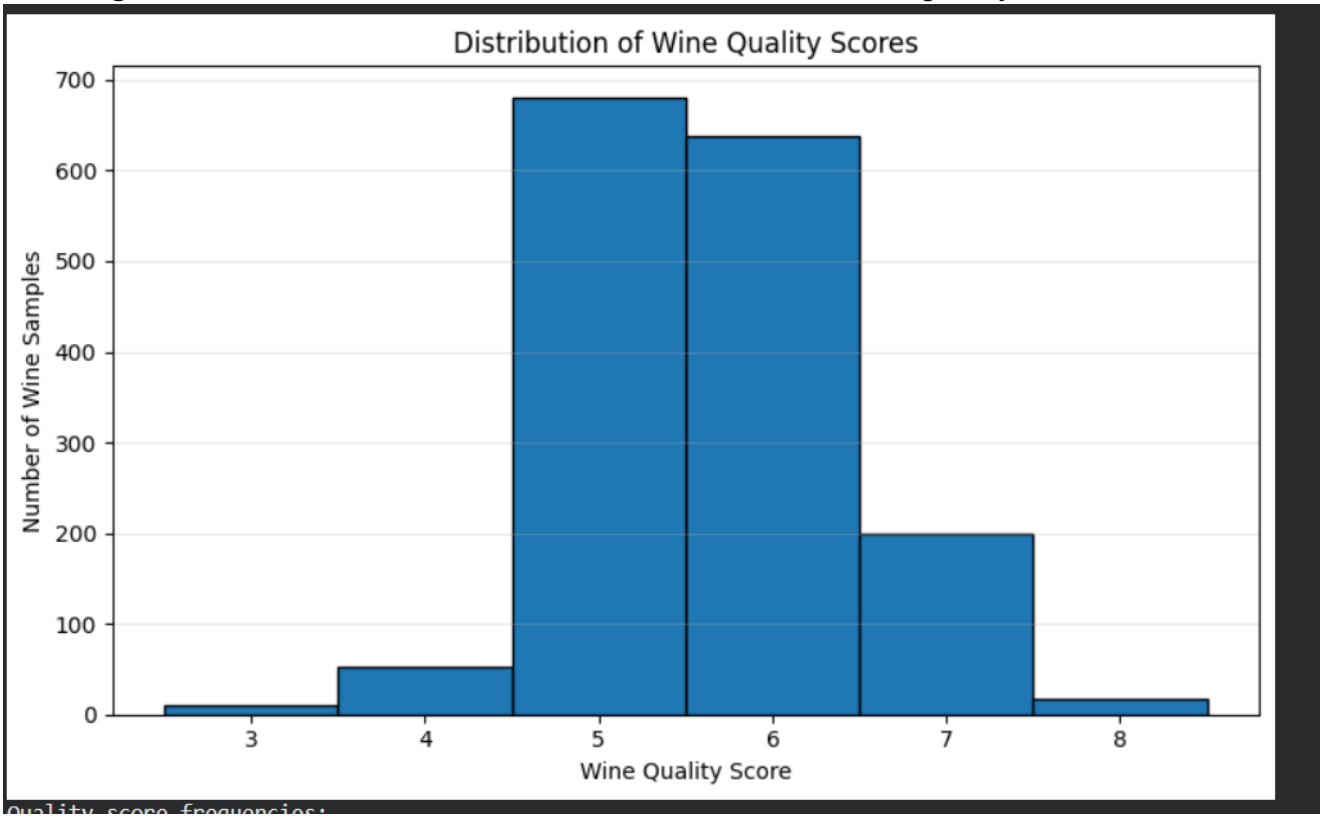
Interpretation

- Alcohol has a moderate positive correlation with wine quality.
- Higher alcohol content tends to be associated with higher quality scores in this dataset.
- Alcohol is an important attribute to consider during quality screening.
- Correlation does not establish causation, and alcohol alone cannot determine wine quality.

8. TASK 2 — EXPLORATORY DATA ANALYSIS

8.1 Task 2(a): Histogram of Wine Quality Scores

A histogram was created to examine the distribution of wine quality scores.



Quality Score Frequency

Quality Score	Number of Samples
3	10
4	53
5	681
6	638
7	199
8	18
Total	**1,599**

Interpretation

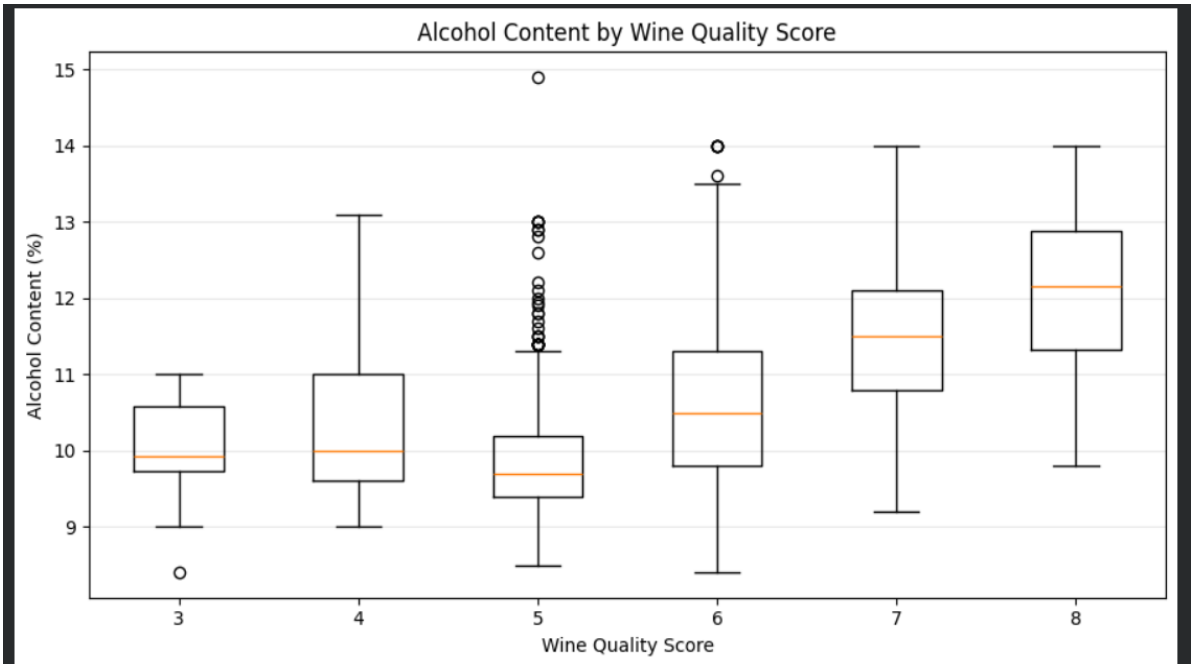
- The most frequent quality score is 5, with 681 samples.
- The next most frequent score is 6, with 638 samples.
- Quality scores 5 and 6 account for approximately 82.49% of the dataset.
- Scores 3 and 8 are relatively uncommon.

- The distribution is concentrated around the middle quality scores.

Conclusion: The distribution is imbalanced, which can make it harder for the model to predict uncommon quality scores reliably.

8.2 Task 2(b): Box Plot of Alcohol by Quality Score

A box plot was generated to compare alcohol content across the different quality scores.

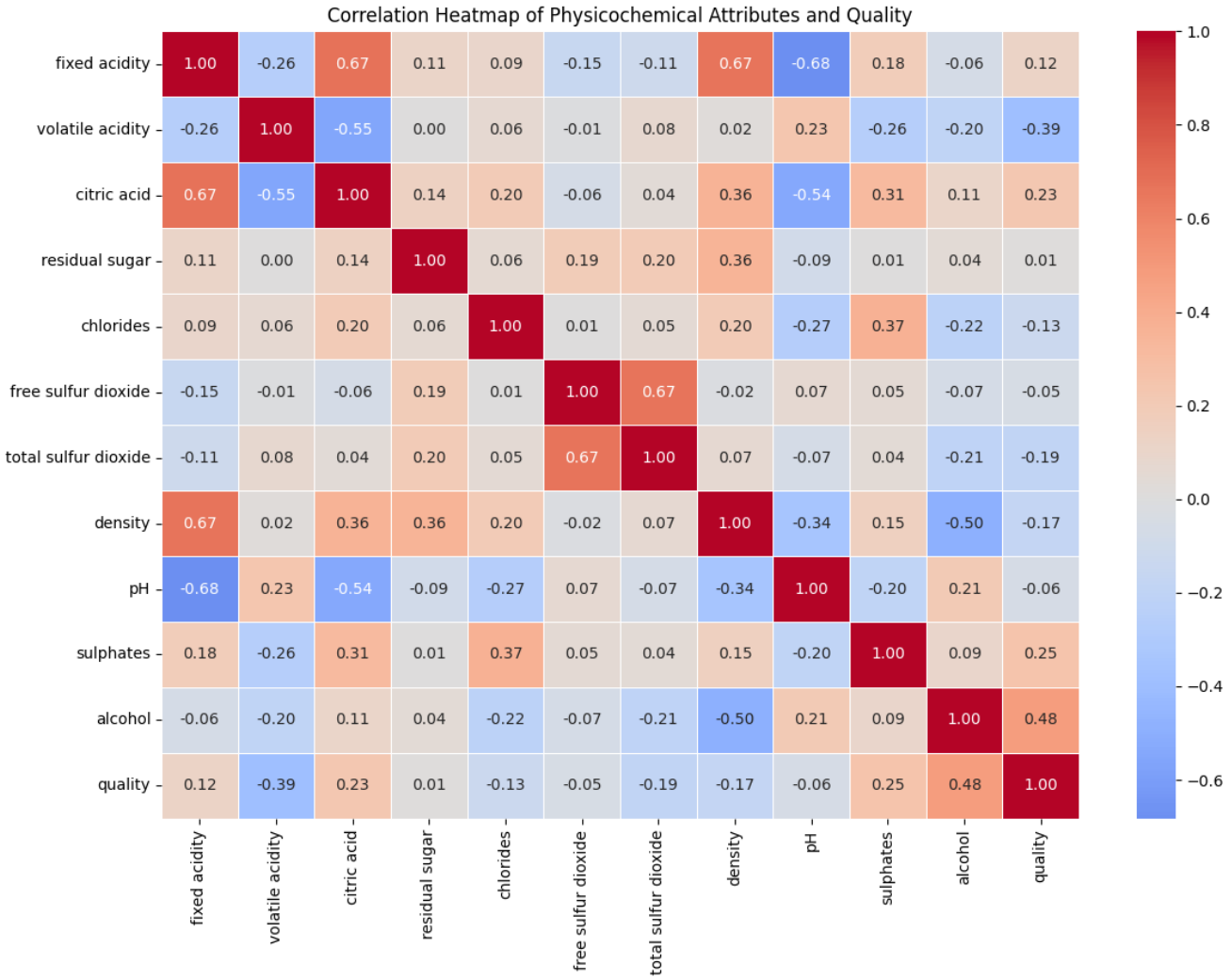
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- The median alcohol content generally increases across higher quality categories.
- The median is approximately 9.7 for quality score 5.
- The median is approximately 10.5 for quality score 6.
- The median is approximately 11.5 for quality score 7.
- The median is approximately 12.15 for quality score 8.
- Some categories contain potential outliers, represented by individual points beyond the whiskers.
- These outliers may be genuine observations and should not automatically be treated as errors.
- Higher quality categories contain fewer observations than the middle categories.

Conclusion: Alcohol content tends to be higher in wine samples with higher quality scores, supporting the positive correlation found in Task 1.

8.3 Task 2(c): Correlation Heatmap

A correlation heatmap was generated to examine relationships between the physicochemical attributes and quality.



- Positive correlation indicates that two variables tend to increase together.
- Negative correlation indicates that one variable tends to decrease as the other increases.
- Alcohol has the strongest positive Pearson correlation with quality.
- Volatile acidity has the strongest negative Pearson correlation with quality.
- Sulphates and citric acid also show positive correlations with quality, but their absolute correlations are weaker than those of alcohol and volatile acidity.
- Relationships between predictors are also relevant because highly related predictors can affect regression coefficient interpretation.

8.4 Task 2(d): Two Attributes Most Strongly Associated With Quality

The attributes were ranked by the absolute value of their Pearson correlations with quality.

Table 3: Strongest Correlations With Quality

Attribute	Pearson Correlation	Direction
Alcohol	+0.4762	Positive
Volatile Acidity	-0.3906	Negative

Interpretation

Alcohol

- Has the strongest absolute Pearson correlation with quality.
- The correlation is positive and moderate.
- Higher alcohol content tends to be associated with higher quality scores.

Volatile Acidity

- Has the second-strongest absolute Pearson correlation with quality.
- The correlation is negative and moderate.
- Higher volatile acidity tends to be associated with lower quality scores.

Conclusion: Alcohol and volatile acidity are the two attributes most strongly associated with quality based on absolute Pearson correlation.

9. TASK 3 — PREDICTIVE MODELLING

9.1 Task 3(a): Multiple Linear Regression

Multiple linear regression was used to predict wine quality using all 11 physicochemical attributes.

Dependent variable:

- Quality

Independent variables:

- Fixed acidity
- Volatile acidity
- Citric acid
- Residual sugar
- Chlorides
- Free sulfur dioxide
- Total sulfur dioxide
- Density
- pH
- Sulphates
- Alcohol

Regression Model

The general model is:

$$\text{Quality} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_{11} X_{11} + \varepsilon$$

Where:

- β_0 represents the intercept.
- β_1 to β_{11} represent the regression coefficients.
- X_1 to X_{11} represent the 11 physicochemical attributes.
- ε represents the unexplained error.

Procedure

- Separated the quality column from the predictor attributes.
- Trained the model on the training subset.
- Used the fitted model to predict scores for the test subset.
- Compared predicted values against actual quality scores.

9.2 Task 3(b): Training and Testing Split

The dataset was divided using an 80:20 split.

Table 4: Data Split

Subset	Number of Samples	Percentage
Training Set	1,279	80%
Testing Set	320	20%
Total	1,599	100%

Procedure

- Used train_test_split() from scikit-learn.
- Set test_size=0.20.
- Set random_state=42.
- Used the training data to fit the regression model.
- Reserved the test data for evaluation.

The fixed random state allows the split to be reproduced when the notebook is run again.

9.3 Task 3(c): Model Evaluation

The model was evaluated using four performance metrics.

- **MAE:** Mean absolute difference between actual and predicted scores.
- **MSE:** Average squared prediction error.
- **RMSE:** Square root of MSE, expressed in quality-score units.
- **R-squared:** Proportion of variation in test-set quality scores explained by the model.

Table 5: Test-Set Performance

Metric	Result
MAE	0.5035
MSE	0.3900
RMSE	0.6245
R-squared	0.4032

- MAE of approximately 0.504 indicates an average absolute prediction error of about half a quality-score point.
- MSE of approximately 0.3900 measures the average squared prediction error.
- RMSE of approximately 0.625 indicates the scale of prediction errors in quality-score units.
- R-squared of approximately 0.4032 indicates that the model explains about 40.32% of the variation in the test-set quality scores.

Conclusion: The regression model identifies useful relationships but has moderate predictive performance. A substantial proportion of the variation in quality remains unexplained.

9.4 Task 3(d): Standardized Regression Coefficients

Standardized regression coefficients were used to compare the relative influence of predictors on a common scale.

The predictors and target were standardised using statistics calculated from the training data.

Table 6: Standardized Regression Coefficients

Attribute	Standardized Coefficient
Alcohol	+0.3678
Volatile Acidity	-0.2225
Sulphates	+0.1820
Total Sulfur Dioxide	-0.1488
Chlorides	-0.1104
pH	-0.0751
Free Sulfur Dioxide	+0.0719
Fixed Acidity	+0.0493
Citric Acid	-0.0341
Density	-0.0238
Residual Sugar	+0.0117

Interpretation

Alcohol

- Has the largest absolute standardized coefficient, approximately +0.3678.
- It is the most influential predictor according to the coefficient magnitude.
- Its positive coefficient indicates a positive conditional association with predicted quality.

Volatile Acidity

- Has the second-largest absolute standardized coefficient, approximately -0.2225.
- Its negative coefficient indicates a negative conditional association with predicted quality, holding the other predictors constant.

Conclusion: Alcohol and volatile acidity are the two most influential predictors according to the standardized regression coefficients.

Multicollinearity Analysis Using VIF

Multicollinearity occurs when predictor variables are strongly related to one another. It can make individual regression coefficients less stable.

Variance Inflation Factor (VIF) was calculated using the training predictors.

Table 7: VIF Results

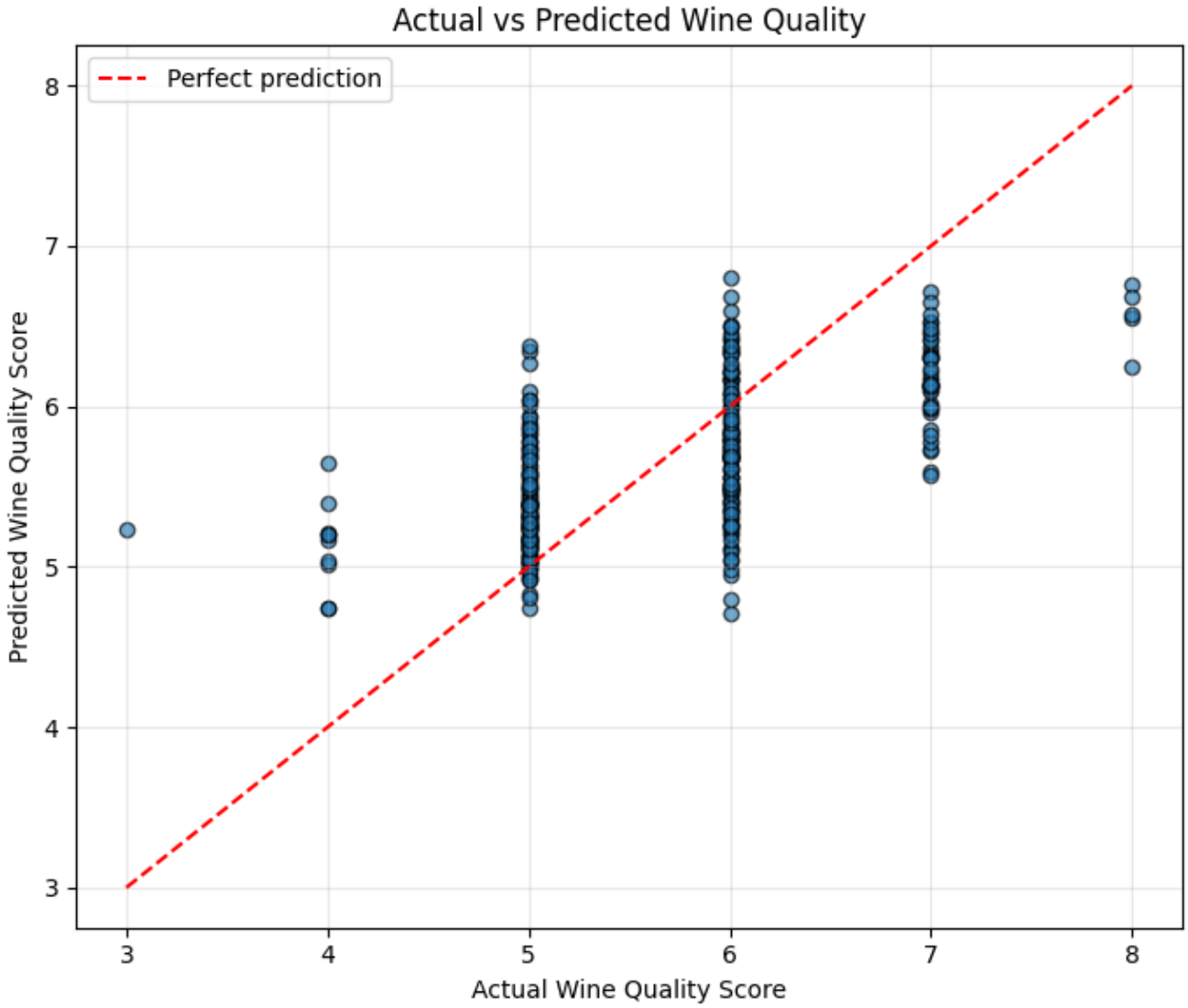
Attribute	VIF
Fixed Acidity	7.375
Density	6.016
pH	3.275
Citric Acid	3.211
Alcohol	2.933

Total Sulfur Dioxide	2.203
Free Sulfur Dioxide	1.983
Volatile Acidity	1.813
Residual Sugar	1.732
Chlorides	1.520
Sulphates	1.456

Interpretation

- Fixed acidity has a VIF of approximately 7.375.
- Density has a VIF of approximately 6.016.
- These values indicate potential multicollinearity concerns because they exceed the commonly used screening threshold of 5.
- The remaining predictors have lower VIF values.
- Therefore, regression coefficients should be interpreted with consideration of relationships among the predictors.
- **9.5 Task 3(e): Actual Versus Predicted Quality**

A scatter plot was generated to compare actual quality scores with predicted quality scores on the test set.



First 10 actual and predicted scores:

	Actual Quality	Predicted Quality
0	**6**	**5.347**
1	**5**	**5.056**
2	**6**	**5.664**
3	**5**	**5.465**
4	**6**	**5.725**
5	**5**	**5.279**
6	**5**	**5.034**
7	**5**	**5.126**
8	**5**	**5.745**
9	**6**	**5.687**

- The dashed diagonal line represents perfect predictions.
- Points closer to the line indicate more accurate predictions.
- Points farther away from the line represent larger prediction errors.
- The model predicts many of the middle quality scores more effectively than the extreme scores.
- The actual scores include values from 3 to 8, while predictions are concentrated more closely around the middle of the range.
- This shows that the model has difficulty predicting uncommon low and high quality scores.

Limitations of the Model

- The R-squared value is approximately 0.4032, so much of the variation in quality is not explained by the model.
- Quality scores are discrete and ordered, while linear regression generates continuous predictions.
- The dataset contains duplicate rows, which may appear in both the training and testing subsets.
- The model uses only the 11 available physicochemical attributes and cannot account for every factor influencing sensory quality.

10. TASK 4 — BUSINESS DECISION-MAKING

10.1 Task 4(a): Recommended Laboratory Measurements

The two recommended measurements are:

1. Alcohol Content

- Pearson correlation with quality: approximately +0.4762.
- Standardized regression coefficient: approximately +0.3678.
- It has the strongest positive correlation with quality.
- Higher alcohol content tends to be associated with higher quality scores in this dataset.

- Alcohol should receive attention during laboratory-based screening.

2. Volatile Acidity

- Pearson correlation with quality: approximately -0.3906.
- Standardized regression coefficient: approximately -0.2225.
- It has the strongest negative correlation with quality.
- Higher volatile acidity tends to be associated with lower quality scores.
- Volatile acidity should be monitored as part of laboratory-based screening.

Recommendation: Prioritise alcohol and volatile acidity when reviewing laboratory measurements. These associations do not prove that changing either attribute will automatically improve wine quality.

10.2 Task 4(b): Can the Model Replace Sensory Evaluation?

Test-Set Results

- MAE = 0.5035
- MSE = 0.3900
- RMSE = 0.6245
- R-squared = 0.4032

Assessment

- The model provides useful information about relationships between laboratory measurements and quality scores.
- The MAE indicates an average absolute prediction error of approximately half a quality-score point.
- The R-squared value indicates that the model explains approximately 40.32% of the variation in test-set quality.
- The model does not capture all factors associated with wine quality.
- The dataset's duplicate observations and uneven quality-score distribution also limit confidence in the evaluation.

Business recommendation: The model should be used as a supporting screening tool, not as a complete replacement for sensory evaluation. Laboratory measurements and predictions can help identify samples requiring closer examination, while trained evaluators should retain responsibility for final quality assessment.

10.3 Task 4(c): Limitations

Limitation 1: Duplicate Observations

- The dataset contains 240 exact duplicate rows.
- The random train-test split may place identical records in both subsets.
- This could lead to data leakage and make test performance appear better than performance on completely new observations.
- A future analysis should investigate duplicate removal and use a suitable validation strategy.

Limitation 2: Uneven Quality-Score Distribution

- Most observations have quality scores of 5 or 6.
- Scores of 3 and 8 are relatively uncommon.
- The model has fewer examples from which to learn the characteristics of extreme quality scores.
- Its predictions may therefore be less reliable for the lowest and highest categories.
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11. CONCLUSION

The case study examined wine quality using statistical analysis, exploratory data analysis and multiple linear regression.

Major Findings

- Residual sugar has the greatest relative variability, with a coefficient of variation of approximately 55.54%.
- Alcohol has the strongest positive Pearson correlation with quality, at approximately +0.4762.
- Volatile acidity has the strongest negative Pearson correlation with quality, at approximately -0.3906.
- Alcohol and volatile acidity are the two leading predictors according to the standardized regression coefficients.
- The regression model achieved an MAE of approximately 0.504, an RMSE of approximately 0.625 and an R-squared value of approximately 0.403.
- VIF analysis indicates potential multicollinearity concerns for fixed acidity and density.

Final Recommendation

- Use alcohol content and volatile acidity as important measurements during laboratory-based quality screening.
- Use the regression model to support decisions and identify samples for further review.
- Do not replace sensory evaluation completely because the model explains only part of the variation in wine quality.
- Validate the model further, particularly with duplicate-free data and an appropriate evaluation strategy, before using it in a real production process.

12. REFERENCES

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